

The closed operator algebra generated by a representation and extra operators

MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget

def

Places a represented algebra and a chosen family of bounded operators in one concrete closed algebra.

Statement

Let A be a unital complex algebra with an additive, multiplicatively reversing involution, and let H be a complex Hilbert space. Write $B(H)$ for its bounded complex-linear operators. Fix a unital complex-linear $*$ -homomorphism $\pi : A \rightarrow B(H)$, that is, a representation satisfying $\pi(a^*) = \pi(a)^*$. For any family $(U_j)_{j \in J}$ in $B(H)$, form the norm-closed unital $*$ -subalgebra generated by $\pi(A)$ and the U_j . This is the concrete target T .

Definition

Put $S = \pi(A) \cup \{U_j : j \in J\}$. Let $\text{alg}_{\mathbb{C}}^*(S)$ denote the smallest unital complex subalgebra of $B(H)$ containing S and closed under adjoints. Define $T = \overline{\text{alg}_{\mathbb{C}}^*(S)}$, where the bar denotes operator-norm closure. Thus T consists of operator-norm limits of finite complex linear combinations of finite products of members of S and their adjoints, including the empty product 1. The adjoint in $B(H)$ and the algebra operations restrict to T .

Assumptions

The algebra A is unital and associative over $\mathbb{C}$, with an additive involution satisfying $(ab)^* = b^*a^*$ and $(a^*)^* = a$. The map π preserves multiplication, the unit, complex scalars and this involution. The space H is complete for its complex inner-product norm. The index set J is arbitrary. No norm on A , faithfulness of π , or conjugate-linearity of the involution on A is assumed.

Conclusion

The construction provides a closed unital star subalgebra $T \subseteq B(H)$ containing the displayed operators. Write $\iota : A \rightarrow T$ for $\iota(a) = \pi(a)$, $g_j \in T$ for the element represented by U_j , and $\rho : T \rightarrow B(H)$ for inclusion. Then $\rho \circ \iota = \pi$ and $\rho(g_j) = U_j$. The inclusion ρ is injective; if π is injective, so is ι . These maps and their injectivity statements are the contextual facts cited below, not additional clauses of the defining declaration.

The closure is taken inside the given operator algebra. The construction does not assert simplicity or irreducibility, and it introduces no universal completion of A . In particular, an arbitrary extra operator U_j need not be unitary.

Main citations

- Exact declaration and proof
- MathlibAnnex.Analysis.CStarAlgebra.generated
- MathlibAnnex.CStarAlgebra.AtomicConstruction.sourceHom

- MathlibAnnex.CStarAlgebra.AtomicConstruction.sourceHom_coe
- MathlibAnnex.CStarAlgebra.AtomicConstruction.generator
- MathlibAnnex.CStarAlgebra.AtomicConstruction.generator_coe
- MathlibAnnex.CStarAlgebra.AtomicConstruction.ambientInclusion
- MathlibAnnex.CStarAlgebra.AtomicConstruction.sourceHom_injective
- MathlibAnnex.CStarAlgebra.AtomicConstruction.ambientInclusion_injective
- MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget.instIsClosed

Lean source signature (exact)

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noncomputable def concreteTarget (pi : A →*a[ $\mathbb{C}$ ] (H →L[ $\mathbb{C}$ ] H))
  (U : J → H →L[ $\mathbb{C}$ ] H) : StarSubalgebra  $\mathbb{C}$  (H →L[ $\mathbb{C}$ ] H) :=
  generated (Set.range pi ∪ Set.range U)
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In the source	Mathematical meaning
$A; H; J$	The given unital associative complex algebra A with additive, multiplicatively reversing involution, a complete complex Hilbert space H , and an arbitrary index set J . No norm or conjugate-linearity of the involution on A is assumed.
$\text{pi} : A \rightarrow_{*a}[\mathbb{C}] (H \rightarrow_L[\mathbb{C}] H)$	The unital complex-linear star homomorphism $\pi : A \rightarrow B(H)$, where $B(H)$ consists of bounded complex-linear operators.
$U : J \rightarrow H \rightarrow_L[\mathbb{C}] H$	The additional operators $(U_j)_{j \in J}$; each is bounded and complex-linear, with no unitary hypothesis.
$\text{StarSubalgebra } \mathbb{C} (H \rightarrow_L[\mathbb{C}] H)$	The output is the concrete unital complex star subalgebra $T \subseteq B(H)$.
$\text{generated } (\text{Set.range pi} \cup \text{Set.range U})$	The full defining RHS forms $T = \overline{\text{alg}_{\mathbb{C}}^*(\pi(A) \cup \{U_j : j \in J\})}$, with operator-norm closure. Set.range pi consists of all $\pi(a)$; Set.range U consists of all U_j . Finite products include the empty product 1, and the generators and their adjoints are allowed.

The displayed complete definition includes the exact RHS of this declaration. Its header-only extraction remains unchanged in the source-binding record.

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▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget

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